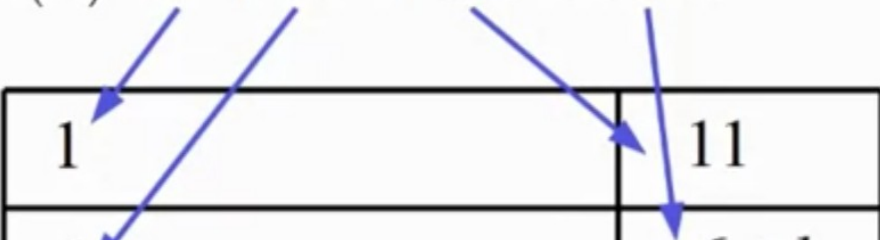


Laboratório de Controle Automático

Restrições sobre parâmetros para a estabilidade

$$\textit{Den}(s)=s^3+6s^2+11s+6+k$$

$$Den(s) = s^3 + 6s^2 + 11s + 6 + k$$



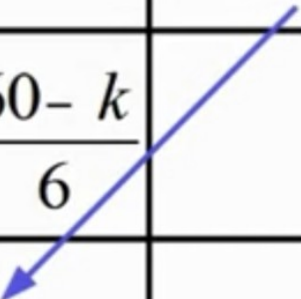
s^3	1	11
s^2	6	$6+k$
s^1		
s^0		

$$Den(s) = s^3 + 6s^2 + 11s + 6 + k$$

s^3	1	11
s^2	6	$6+k$
s^1	$\frac{6 \cdot 11 - 6 - k}{6} = \frac{60 - k}{6}$	
s^0		

$$Den(s) = s^3 + 6s^2 + 11s + 6 + k$$

s^3	1	11
s^2	6	$6+k$
s^1	$\frac{6 \cdot 11 - 6 \cdot k}{6} = \frac{60 - k}{6}$	
s^0	$6+k$	



$$\frac{60 - k}{6} > 0$$

$$k < 60$$

$$6 + k > 0$$

$$k > -6$$

$$Den(s) = s^3 + 6s^2 + 11s + 66$$

$$Den(s) = (s + 6)(s + \sqrt{11}i)(s - \sqrt{11}i)$$

$$Den(s) = (s + 6)(s^2 - (\sqrt{11})^2(i)^2)$$

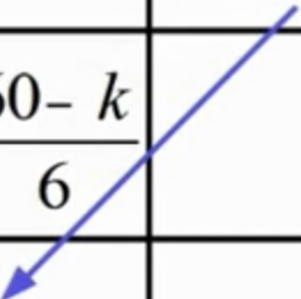
$$Den(s) = (s + 6)(s^2 - 11(-1)) = (s + 6)(s^2 + 11)$$

$$\textit{Den}(s)=s^3+6s^2+11s$$

$$\textit{Den}(s)=s(s^2+6s+11)$$

$$Den(s) = s^3 + 6s^2 + 11s + 6 + k$$

s^3	1	11
s^2	6	$6+k$
s^1	$\frac{6 \cdot 11 - 6 \cdot k}{6} = \frac{60 - k}{6}$	
s^0	$6+k$	

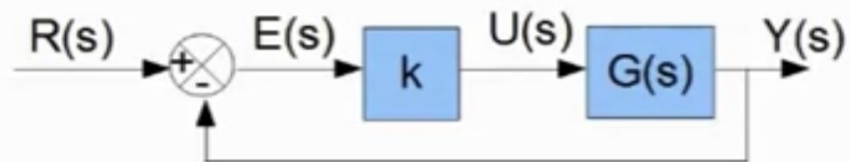


$$\frac{60 - k}{6} > 0$$

$$k < 60$$

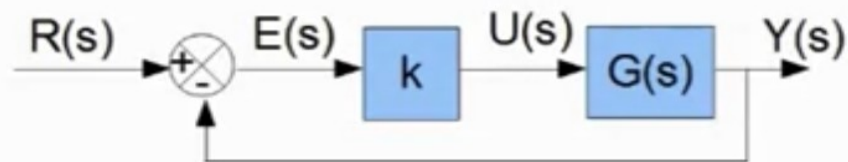
$$6 + k > 0$$

$$k > -6$$



$$G(s) = \frac{1}{s^2 + 2s - 3}$$

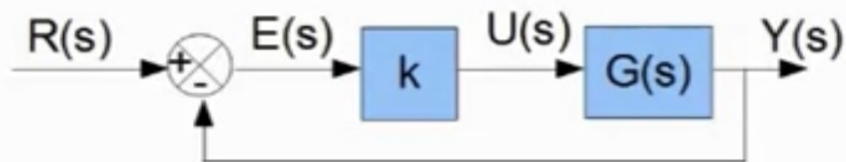
$$T(s) = \frac{k G(s)}{1 + k G(s)}$$



$$G(s) = \frac{1}{s^2 + 2s - 3}$$

$$T(s) = \frac{k}{s^2 + 2s - 3 + k}$$

s^2	1	$-3 + k$
s^1	2	
s^0	$-3 + k$	

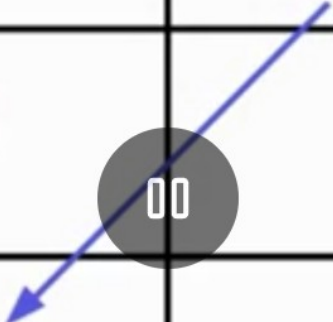


$$G(s) = \frac{10}{s^3 + 8s^2 + 12s + 10} \quad T(s) = \frac{k G(s)}{1 + k G(s)}$$

$$T(s) = \frac{10k}{s^3 + 8s^2 + 12s + 10 + 10k}$$

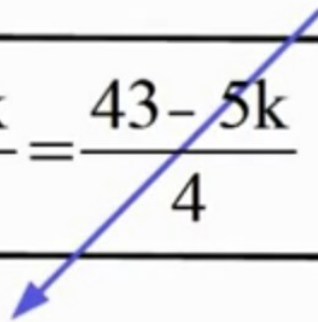
$$DenG(s) = s^3 + 8s^2 + 12s + 10$$

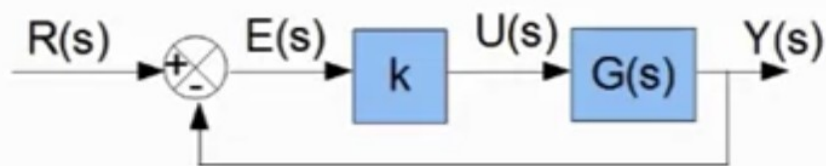
s^3	1	12
s^2	8	10
s^1	$\frac{8 \cdot 12 - 10}{8} = \frac{43}{4}$	
s^0	10	



$$DenT(s) = s^3 + 8s^2 + 12s + 10 + 10k$$

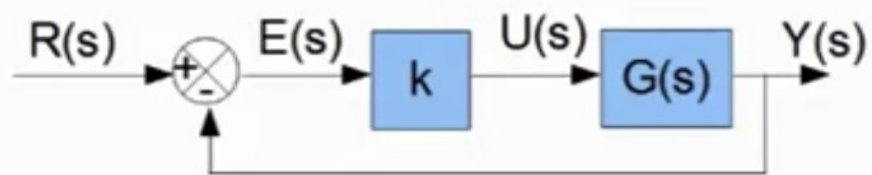
s^3	1	12
s^2	8	$10 + 10k$
s^1	$\frac{8 \cdot 12 - 10 - 10k}{8} = \frac{43 - 5k}{4}$	
s^0	$10 + 10k$	





$$G(s) = \frac{10}{s^3 + 8s^2 + 12s + 10}$$

$$T(s) = \frac{10k}{s^3 + 8s^2 + 12s + 10 + 10k}$$

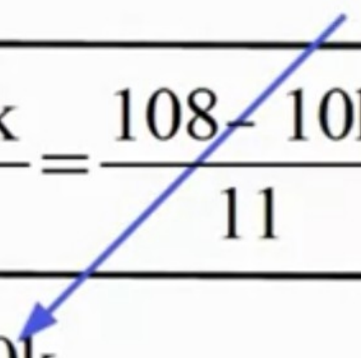


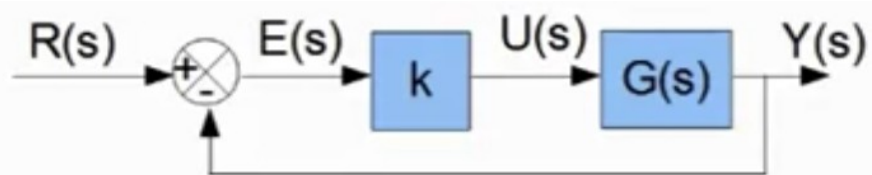
$$G(s) = \frac{10}{s^3 + 11s^2 + 8s - 20}$$

$$T(s) = \frac{10k}{s^3 + 11s^2 + 8s - 20 + 10k}$$

$$DenT(s) = s^3 + 11s^2 + 8s - 20 + 10k$$

s^3	1	8
s^2	11	$-20 + 10k$
s^1	$\frac{11 \cdot 8 + 20 - 10k}{11} = \frac{108 - 10k}{11}$	
s^0	$-20 + 10k$	





$$G(s) = \frac{10}{s^3 + 11s^2 + 8s - 20}$$

$$T(s) = \frac{10k}{s^3 + 11s^2 + 8s - 20 + 10k}$$

$$T(s) = \frac{k}{s^2 - s - 2 + k_i s + k}$$

$$DenT(s) = s^2 - s - 2 + k_t s + k$$

s^2	1	$-2 + k$
s^1	$k_t - 1$	
s^0	$-2 + k$	

$$T(s) = \frac{as + b}{s^3 + 3s^2 + 2s - 1 + as + b}$$

$$DenT(s) = s^3 + 3s^2 + 2s - 1 + as + b$$

s^3	1	$2 + a$
s^2	3	$-1 + b$
s^1	$\frac{6 + 3a + 1 - b}{3} = \frac{7 + 3a - b}{3}$	
s^0	$-1 + b$	

$$T(s) = \frac{k(s+a)}{s^3 + 3s^2 + 2s + k(s+a)}$$

$$DenT(s) = s^3 + 3s^2 + 2s + k(s + a)$$

s^3	1	$2 + k$
s^2	3	$k a$
s^1	$\frac{6 + 3k - k a}{3}$	
s^0	$k a$	